

ER diagram to Relational schema

```
CREATE TABLE TimePeriod (  
    FL_DATE DATE PRIMARY KEY,  
    YEAR SMALLINT NOT NULL,  
    MONTH VARCHAR(20) NOT NULL,  
    DAY_OF_MONTH TINYINT NOT NULL,  
    DAY_OF_WEEK VARCHAR(20) NOT NULL,  
    QUARTER VARCHAR(20) NOT NULL  
);
```

```
CREATE TABLE Airlines(  
    OP_CARRIER_AIRLINE_ID INT PRIMARY KEY,  
    OP_UNIQUE_CARRIER VARCHAR(100) NOT NULL  
);
```

```
CREATE TABLE Airports(  
    AIRPORT_ID INT PRIMARY KEY,  
    IATA_CODE VARCHAR(20) NOT NULL,  
    CITY_NAME VARCHAR(80) NOT NULL,  
    STATE_ABR VARCHAR(20) NOT NULL,  
    FOREIGN KEY (STATE_ABR) REFERENCES State(STATE_ABR)  
);
```

```
CREATE TABLE Flights (  
    FLIGHT_ID INT PRIMARY KEY,  
    OP_CARRIER_FL_NUM INT NOT NULL,  
    TAIL_NUM VARCHAR(20) NULL,  
    OP_CARRIER_AIRLINE_ID INT NOT NULL,
```

```
ROUTE_ID INT NOT NULL,  
  
FL_DATE DATE NOT NULL,  
  
CANCELLED TINYINT NOT NULL,  
  
DIVERTED TINYINT NOT NULL,  
  
FOREIGN KEY (OP_CARRIER_AIRLINE_ID) REFERENCES  
Airlines(OP_CARRIER_AIRLINE_ID),  
  
FOREIGN KEY (ROUTE_ID) REFERENCES Routes(ROUTE_ID),  
  
FOREIGN KEY (FL_DATE) REFERENCES TimePeriod(FL_DATE)  
  
);
```

```
CREATE TABLE FlightPerformance (  
  
    FLIGHT_ID INT PRIMARY KEY,  
  
    CRS_DEP_TIME TIME NOT NULL,  
  
    DEP_TIME TIME,  
  
    DEP_DELAY INT,  
  
    DEP_DELAY_NEW INT,  
  
    DEP_DEL15 TINYINT,  
  
    TAXI_OUT SMALLINT,  
  
    WHEELS_OFF TIME,  
  
    WHEELS_ON TIME,  
  
    TAXI_IN SMALLINT,  
  
    CRS_ARR_TIME TIME,  
  
    ARR_TIME TIME,  
  
    ARR_DELAY INT,  
  
    ARR_DELAY_NEW INT,
```

```
ARR_DEL15 TINYINT,  
  
CRS_ELAPSED_TIME SMALLINT,  
  
ACTUAL_ELAPSED_TIME SMALLINT,  
  
AIR_TIME SMALLINT,  
  
CANCELLATION_CODE CHAR(50),  
  
DIV_AIRPORT_LANDINGS TINYINT,  
  
DIV_REACHED_DEST TINYINT,  
  
DIV_ACTUAL_ELAPSED_TIME TINYINT,  
  
DIV_ARR_DELAY INT,  
  
DIV_DISTANCE INT,  
  
FOREIGN KEY (FLIGHT_ID) REFERENCES Flights(FLIGHT_ID),  
  
FOREIGN KEY (CANCELLATION_CODE) REFERENCES  
CancellationReason(CANCELLATION_CODE)  
);
```

```
CREATE TABLE Routes (  
  
    ROUTE_ID INT PRIMARY KEY AUTO_INCREMENT,  
  
    ORIGIN_AIRPORT_ID INT NOT NULL,  
  
    DEST_AIRPORT_ID INT NOT NULL,  
  
    DISTANCE INT NOT NULL,  
  
    FOREIGN KEY (ORIGIN_AIRPORT_ID) REFERENCES Airports(AIRPORT_ID),  
  
    FOREIGN KEY (DEST_AIRPORT_ID) REFERENCES Airports(AIRPORT_ID)  
);
```

```
CREATE TABLE FlightDelay (  
  
    DELAY_ID TINYINT NOT NULL,
```

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    FLIGHT_ID INT NOT NULL,

    MINUTES INT,

    PRIMARY KEY (DELAY_ID, FLIGHT_ID),

    FOREIGN KEY (DELAY_ID) REFERENCES DelayType(DELAY_ID),

    FOREIGN KEY (FLIGHT_ID) REFERENCES Flights(FLIGHT_ID)

);

CREATE TABLE FlightDiversion (

    FLIGHT_ID INT NOT NULL,

    DIVERSION_NUMBER TINYINT NOT NULL,

    DIV_AIRPORT_ID INT NOT NULL,

    DIV_TOTAL_GTIME SMALLINT NULL,

    PRIMARY KEY (FLIGHT_ID, DIVERSION_NUMBER),

    FOREIGN KEY (FLIGHT_ID) REFERENCES Flights(FLIGHT_ID),

    FOREIGN KEY (DIV_AIRPORT_ID) REFERENCES Airports(AIRPORT_ID)

);

```

```

CREATE TABLE State (

    STATE_ABR CHAR(2) PRIMARY KEY,

    STATE_NM VARCHAR(50) NOT NULL

);

```

```

CREATE TABLE AirlineNames (

    OP_CARRIER_AIRLINE_ID INT PRIMARY KEY,

    NAME VARCHAR(150) NOT NULL

);

```

```

CREATE TABLE AirportNames (

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AIRPORT_ID INT PRIMARY KEY,  
NAME VARCHAR(100) NOT NULL  
);  
CREATE TABLE DelayType (  
    DELAY_ID TINYINT PRIMARY KEY,  
    NAME VARCHAR(50) NOT NULL  
);  
CREATE TABLE CancellationReason (  
    CANCELLATION_CODE CHAR(1) PRIMARY KEY,  
    DESCRIPTION VARCHAR(50) NOT NULL  
);
```